

Since keeping volatility low is just as important as obtaining high returns, I do think it makes more sense to use intermediate-term government bonds than to use corporate bonds. The updated results for the Trinity study provided in Exhibit 4.5 follow Mr. Bengen's choice and include the assumption that withdrawals are made at the start of the year rather than the end of the year.

William Bengen's research gave us the SAFEMAX, which he defined as the highest sustainable withdrawal rate in the worst-case scenario from rolling periods of the historical data. The Trinity study went a step further by tallying up the percentage of times that withdrawal rates fell below or above certain levels. They calculated these portfolio success rates for different withdrawal rates, for different time horizons, and for different asset allocations.

For instance, what is the success rate for a 5 percent inflation-adjusted withdrawal rate over a thirty-year period with a 50/50 asset allocation? To answer, note that there are sixty-two rolling thirty-year periods starting between 1926 and 2016. These rolling thirty-year periods begin in the years 1926 through 1987. We do not yet fully know the thirty-year results for more recent starting periods. Of these sixty-two rolling periods, we can refer back to Exhibit 4.1 and count the number of times that the historical surviving withdrawal rate was at least 5 percent. The answer is forty-three times. That means the portfolio success rate is $100 \times 43 / 62 = 69\%$. This is shown in the exhibit, along with a variety of other permutations.

Exhibit 4.5

Portfolio Success Rates

Inflation-Adjusted Withdrawals

For Various Withdrawal Rates, Asset Allocations, and Retirement Durations

Using Ibbotson's Stocks, Bonds, Bills, and Inflation Data, 1926–2016, S&P 500 and Intermediate-Term Government Bonds